

day 38 prep

first we need a way to produce something like a square wave or a saw-tooth wave.

we also need to go through the math of fourier transforms. then go into fast fourier transforms, then on to doing some fourier transforms of simple things and then more complex things.

the mod function works great here and it is a real pro tip.

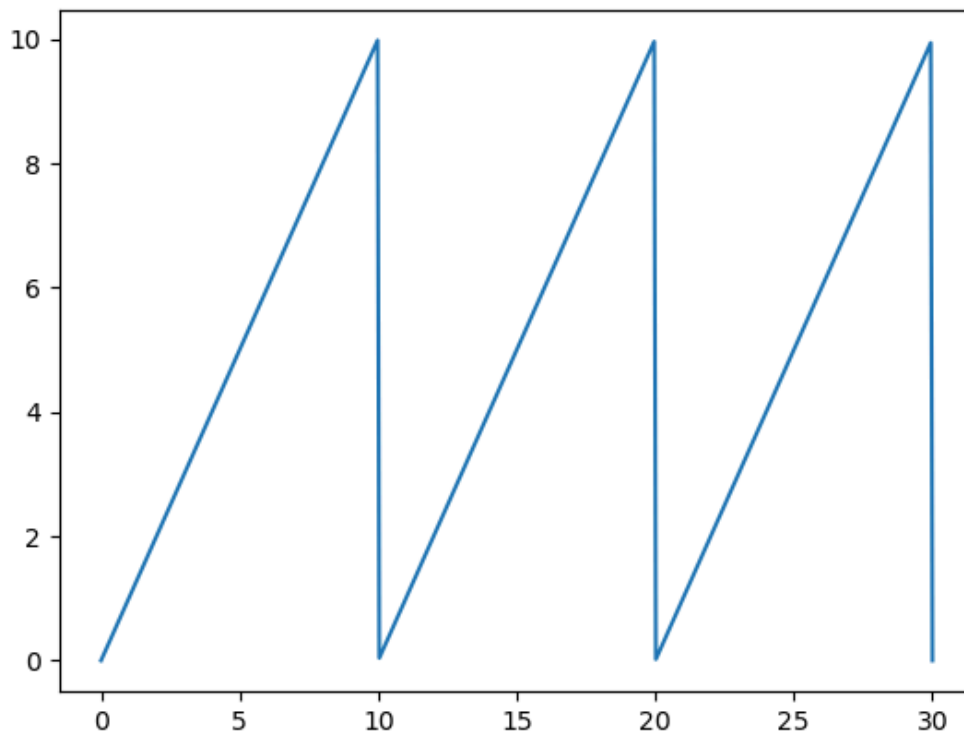
```
In [24]: x = np.linspace(0,10,11)
y = x

L = 10
x = np.linspace(0, 3*L, 500)  # show three periods bc it looks cooler
y = (x % L)                   # periodic sawtooth

fig0,ax0=plt.subplots()
ax0.plot(x, y)
```

```
Out[24]: [<matplotlib.lines.Line2D at 0x7099115640e0>]
```

Figure

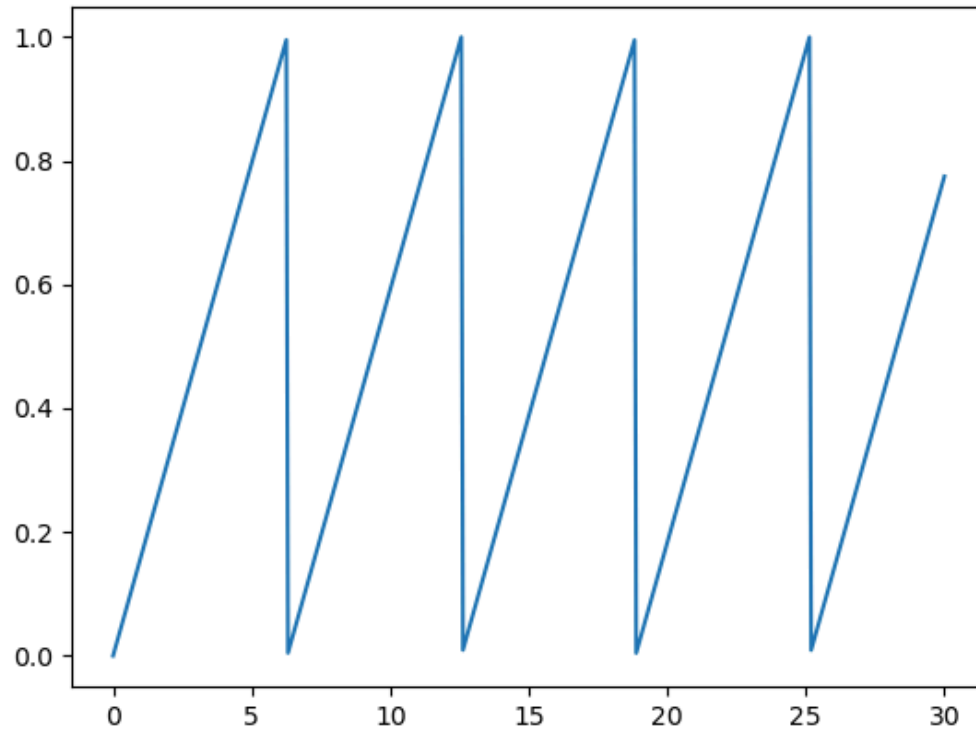


so how can i make a sawtooth of that quickly.

```
In [25]: def saw(x, L=2*np.pi):  
         return (x % L)/L # normalized to [0,1]  
  
         fig1, ax1 = plt.subplots()  
         ax1.plot(x,saw(x))
```

Out[25]: [

Figure



In []: